

Name: _____ Date: _____

Period: _____

Parametrically Defined Circles and Lines Practice

1. A circular fountain has radius 6 m, centered at the point $(-1, 3)$. A water jet starts at $(-1 + 6, 3) = (5, 3)$ and rotates counterclockwise.

(a) Write parametric equations for the jet's position at time t .

(b) Find the jet's position when $t = \pi/2$ and when $t = \pi$.

2. Consider the circular path $x(t) = -2 + 5 \cos t$, $y(t) = 4 + 5 \sin t$, $0 \leq t \leq 2\pi$.

(a) State the center and radius of the circle.

(b) Find all values of t in the domain where $x(t) = -2$. What are the coordinates of the circle at those values?

3. Write the parametric equations for a *clockwise* circle of radius 7 centered at $(2, -3)$.

At $t = \pi/2$, where is the point? Show the computation.

4. A hiker walks from campsite $A = (1, 4)$ to viewpoint $B = (9, 0)$ in a straight line.

(a) Write parametric equations for the hiker's position for $0 \leq t \leq 1$.

(b) Find the hiker's position at $t = 0.25$ and at $t = 0.75$.

(c) At what parameter value t does the hiker's path cross the x -axis (where $y = 0$)? Show your algebra.

AP-Style Practice

5. The parametric equations $x(t) = 3 + 4 \cos t$ and $y(t) = -2 + 4 \sin t$, $0 \leq t \leq 2\pi$, describe the position of a point moving along a circle. Which of the following is the y -coordinate of the point when $x(t) = 3$?

(A) 2

(B) -6

(C) 2 and -6

(D) -2

Extension optional

Extension (optional): The point $(x(t), y(t)) = (h + r \cos t, k + r \sin t)$ lies on a circle. Show algebraically (by eliminating t) that $(x - h)^2 + (y - k)^2 = r^2$. *Hint: isolate $\cos t$ and $\sin t$ first, then use $\cos^2 t + \sin^2 t = 1$.*
